

6 Digit Nixie Clock "64Bit" - Driver Board

64Bit comes from the fact that this clock is driven with shift registers. Between the 6 nixie digits, and indicators, 64 bits are sent from the microcontroller to the shift registers to create the display output

This is an STM32F446 based project.

The clock time keeping is handled via a Real Time Clock (RTC) communicated with via I2C. The STM32 also has an on-board RTC that can be utilized. Both RTCs have a battery backup via a replaceable 2032 coin cell.

The interface with the clock is via a rotary encoder with a push button. Interfacing with remote capacitive buttons is possible via connectors on the bottom of the Driver Board.

There is an ESP32-C6 module driving a full-color LCD screen. The ESP32 features wifi and bluetooth allowing for automatic time setting and smart home integration. Communication with the STM32 is via UART.

This clock was designed with safety in mind. There is a dedicated safety circuit that cuts the input and output connections of the high voltage power supply via relays. There are connectors for interfacing external spring pin connections to the cover of the clock, ensuring the high voltage power supply is not on while the lid is off. The safety circuit features dual channels and feedback to the microcontroller. With the spring pin feature, it will provide isolation from the high voltage, even in an event where the microcontroller experiences an error.

The power design of the clock relays on simple components. There are two inputs available to power the clock. A USB-C port and a 12V barrel jack. A boost converter utilizing XL6009 converts (boost) the incoming voltage to 12VDC. Then an LDO creates 5V then another LDO, in series, creates 3.3V. The high voltage power supply typically is set to output 170VDC to 180VDC. It is supplied by a power supply module that sources 12V power.

This driver board interface with the display board PCB via right angle board-to-board connectors. Mounted to the back of the display board is an LED board to provide RGB backlighting to each IN-12 Nixie Tubes.

The circuit boards will be mounted to a 3D printed base via brass standoffs. Overtop of the PCBs will be a glass top. The glass top will have sections where copper tape is applied. Spring pins contact the copper tape to pass connections for the safety circuit.

STM32 MCU

I2C

Interface

Safety Circuit

Power

Connectors

Connectors 2

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Sheet: /

File: 64 Bit - Main Board.kicad_sch

Title: Project Overview

Size: A4

Date: 2025-12-01

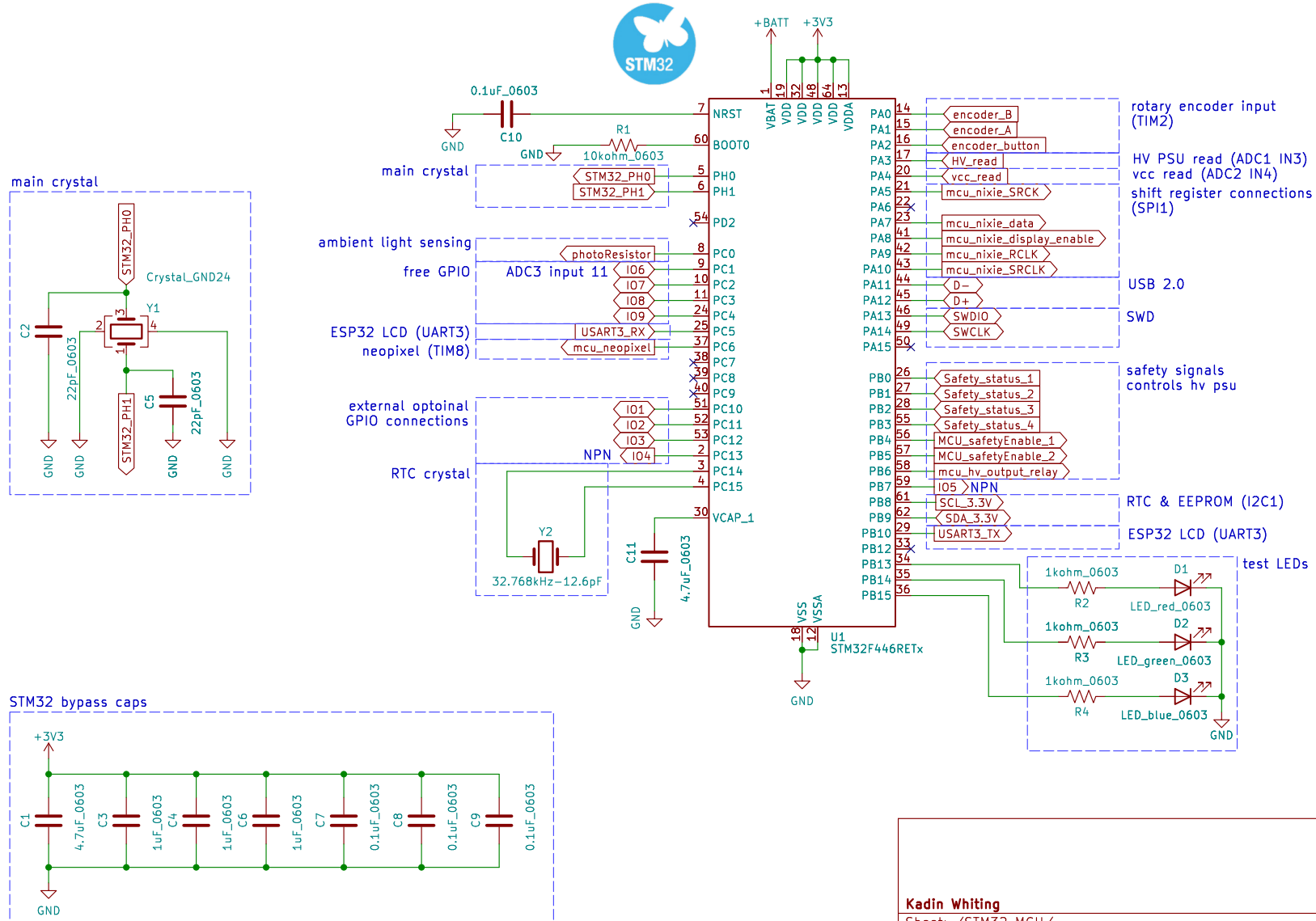
Rev: 1

KiCad E.D.A. 9.0.6

Id: 1/9

STM32F446 Processor ARM M4
168MHz

Level Shifting



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Sheet: /STM32 MCU/

File: STM32 MCU.kicad_sch

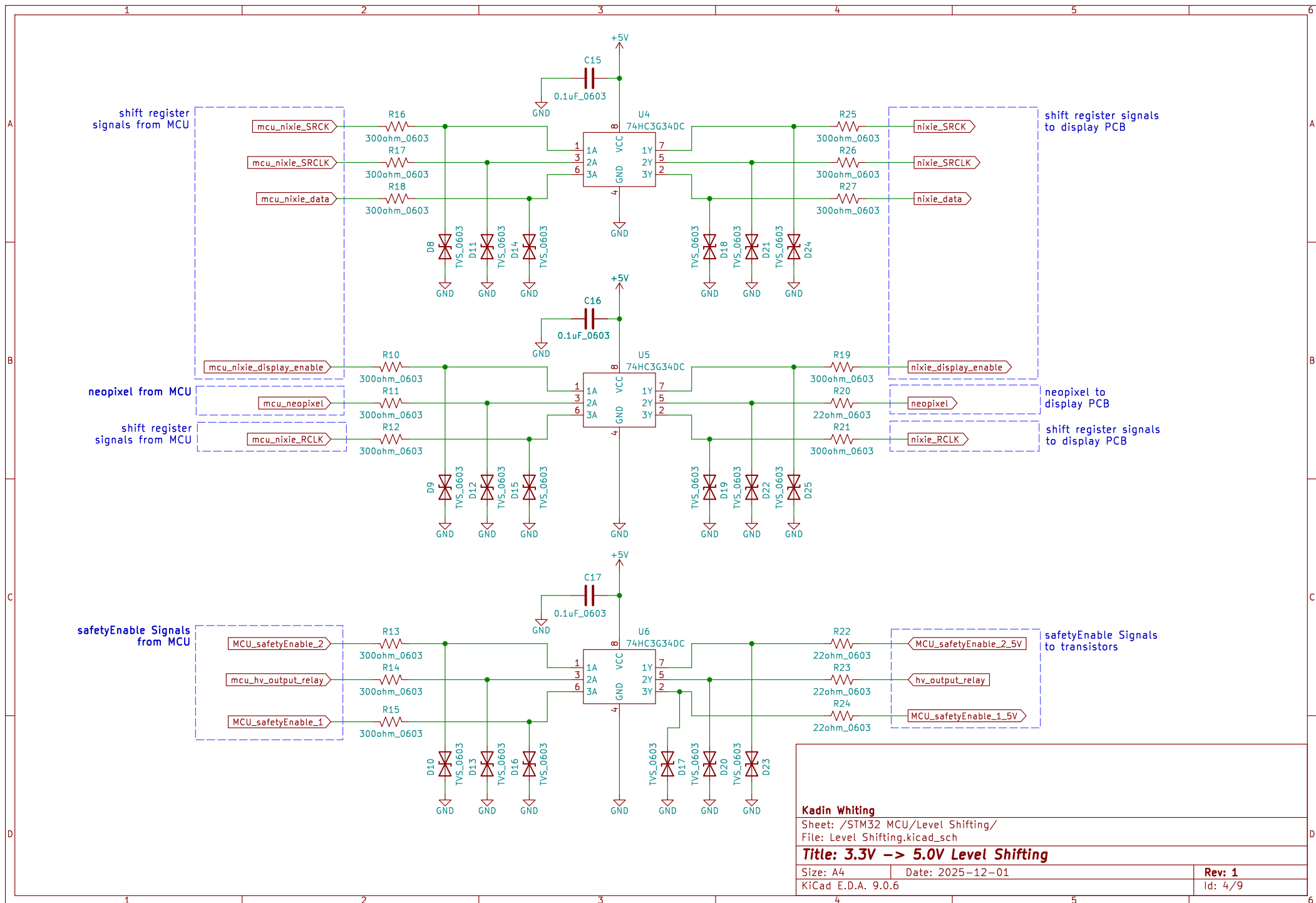
Title: Microcontroller

Size: A4 Date: 2025-12-01

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Rev: 1

Id: 2/9



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Sheet: /STM32 MCU/Level Shifting/
File: Level Shifting.kicad_sch

Title: 3.3V -> 5.0V Level Shifting

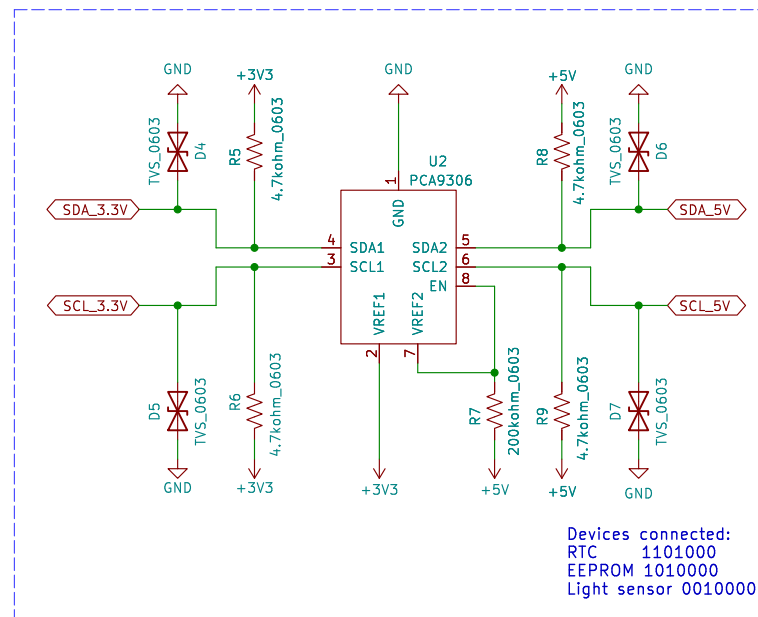
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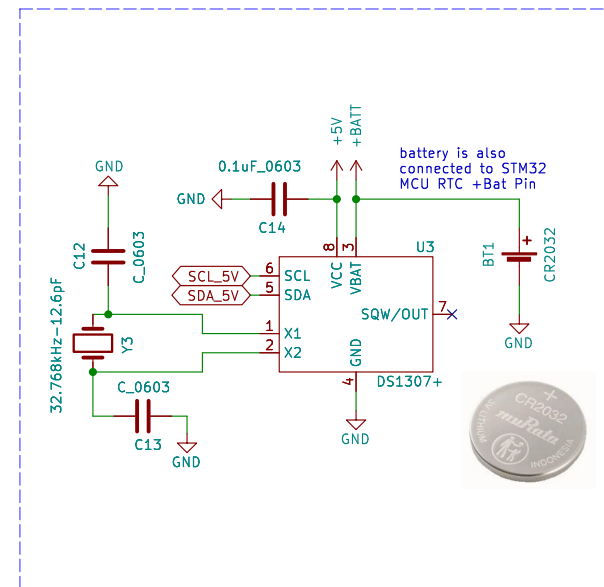
Rev: 1

Id: 4/9

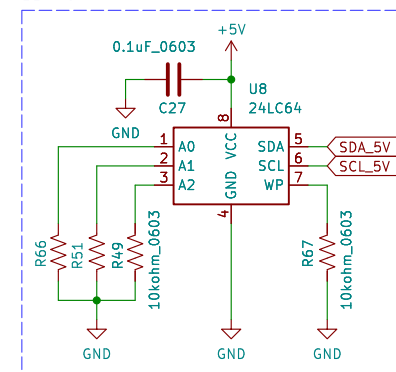
3.3V to 5V I2C level shifter



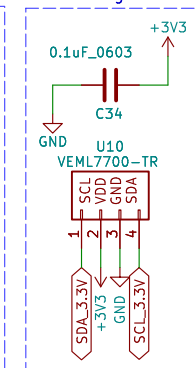
Real Time Clock



EEPROM



ambient light sensor



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Sheet: /I2C/
 File: I2C.kicad_sch

Title: RTC & RTC Level Shifting

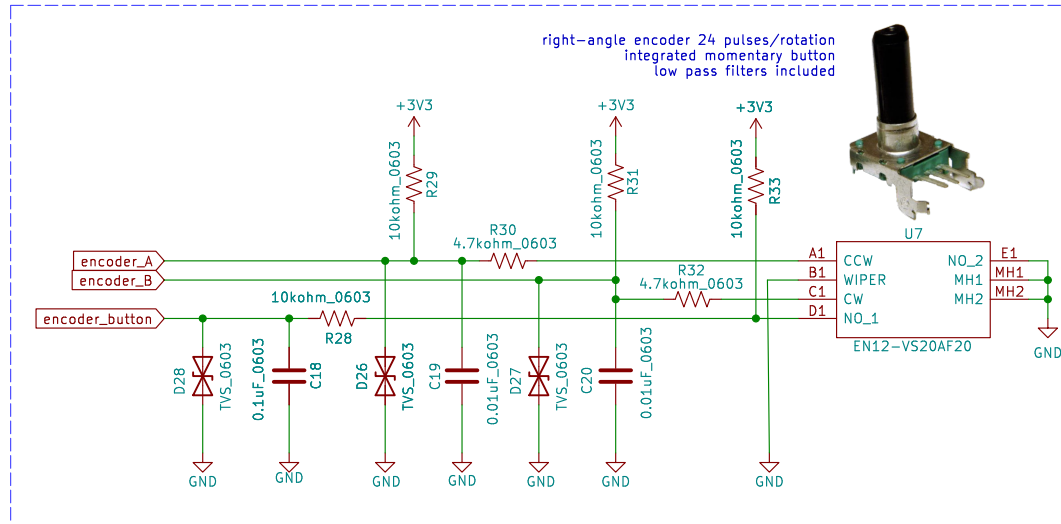
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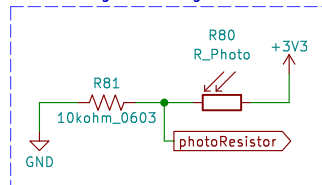
Rev: 1

Id: 3/9

rotary encoder with push button



photoresistor ambient light sensing



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Sheet: /Interface/
File: Interface.kicad_sch

Title: Interface – Rotary Encoder

Size: A4 Date: 2025-12-01

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Rev: 1

Id: 5/9

two outputs from MCU
to enable safety loop

**P-Channel not suitable try:
SQ2389PES-T1_GE3
default gate voltage must be 12V**

safety contacts for sensing if the lid is installed
feedback to MCU for each contact

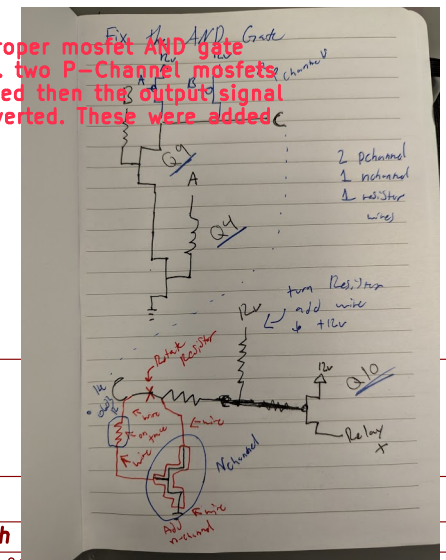
loop through
display PCB

enables HV PSU input relay

controls output of HV power supply

this relay is to allow the
hv power supply voltage to
charge up before applying
the load
it is redundant protection
considering the output gate
shift registers can be set
to all off

this is not a proper mosfet
implementation. two P-Channel
mosfets
need to be added then the output signal
needs to be inverted. These were added
via wires



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Sheet: /Safety Circuit/

File: Safety_Circuit.kicad_sch

Title: Safety Circuit - High

Size: A4

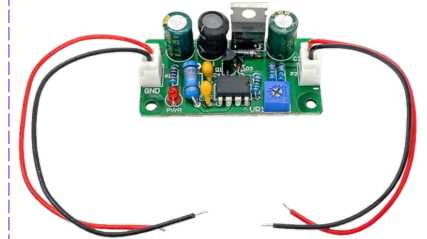
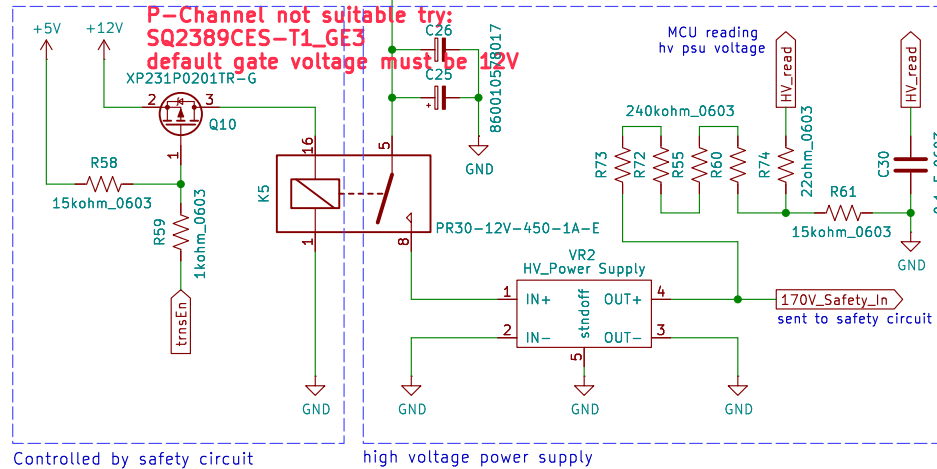
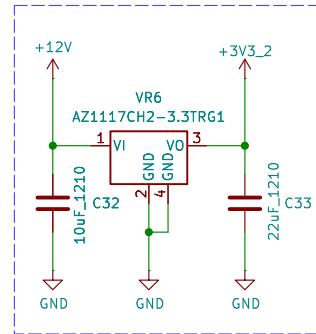
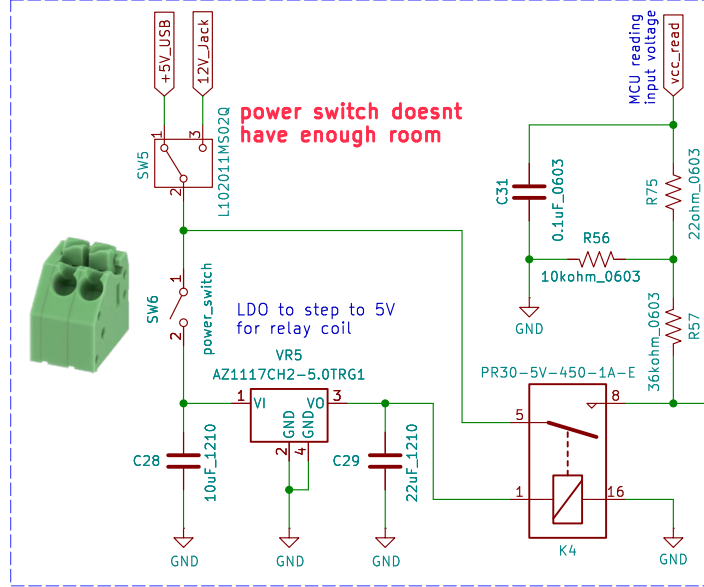
Date: 2025-12-01

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Rev: 1

Id: 6/9

power selection and power switch



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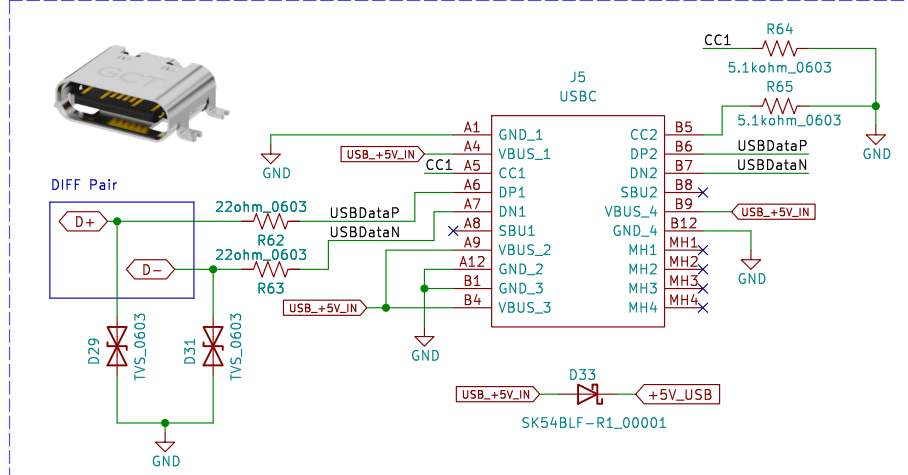
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File: Power.kicad_sch

Title: Power

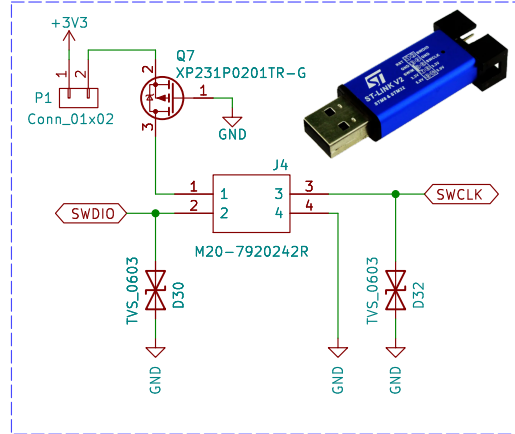
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KiCad E.D.A. 9.0.6

Rev: 1
Id: 7/9

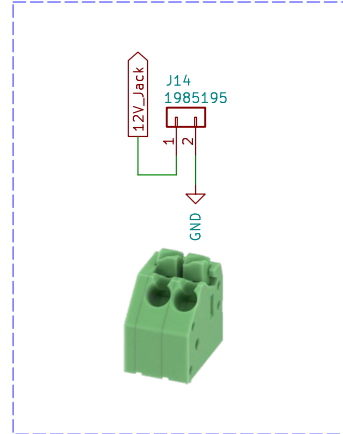
USB-C (2.0 Device Only)



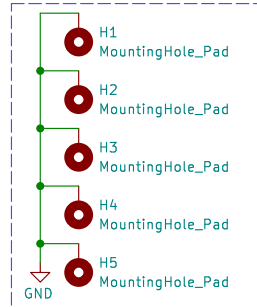
Programming Header



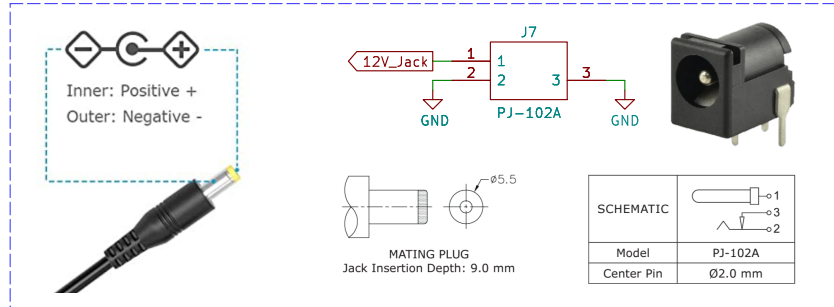
12VDC backup input



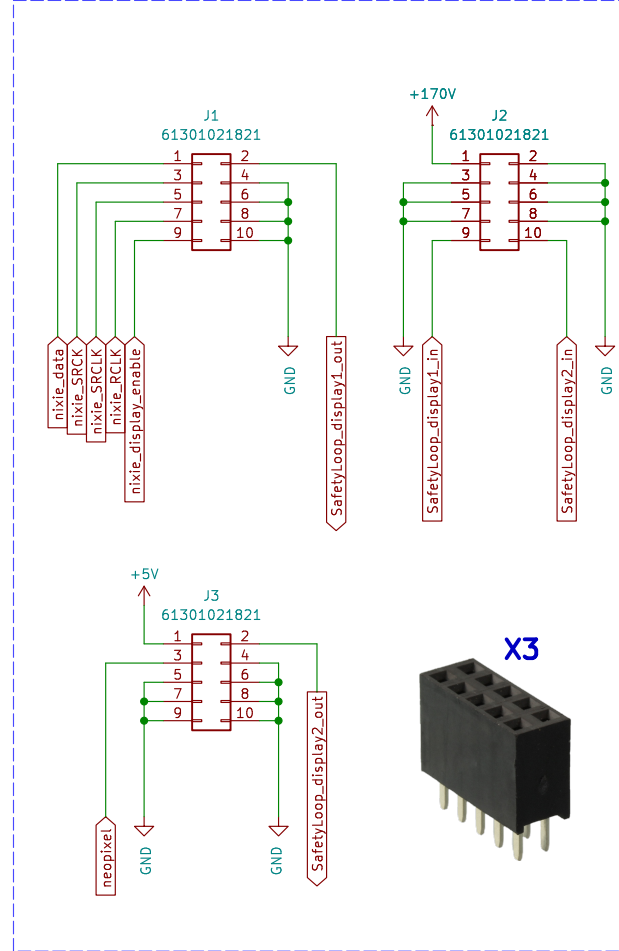
M3 mounting holes



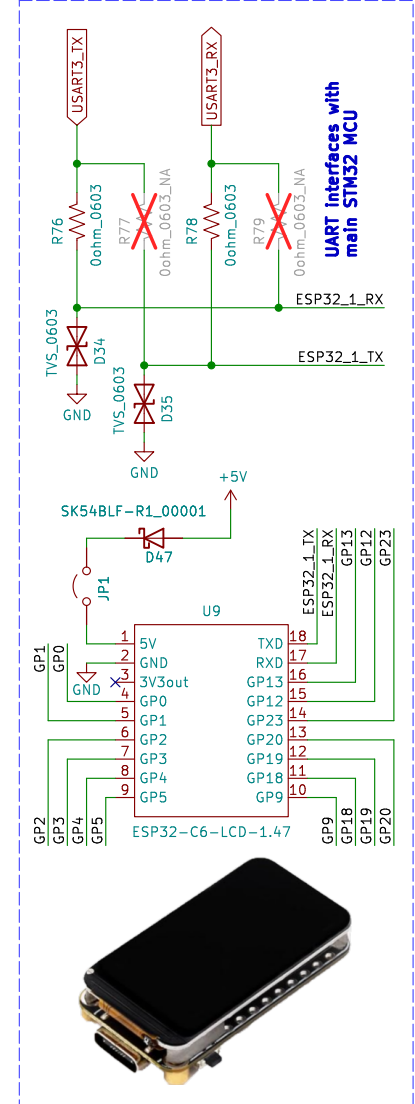
12VDC power input



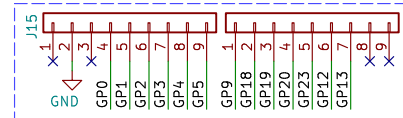
connections to driver board



ESP32-C6 1.47" 172x320 LCD



ESP32 Breakout Connections



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Sheet: /Connectors/

File: Connectors.kicad_sch

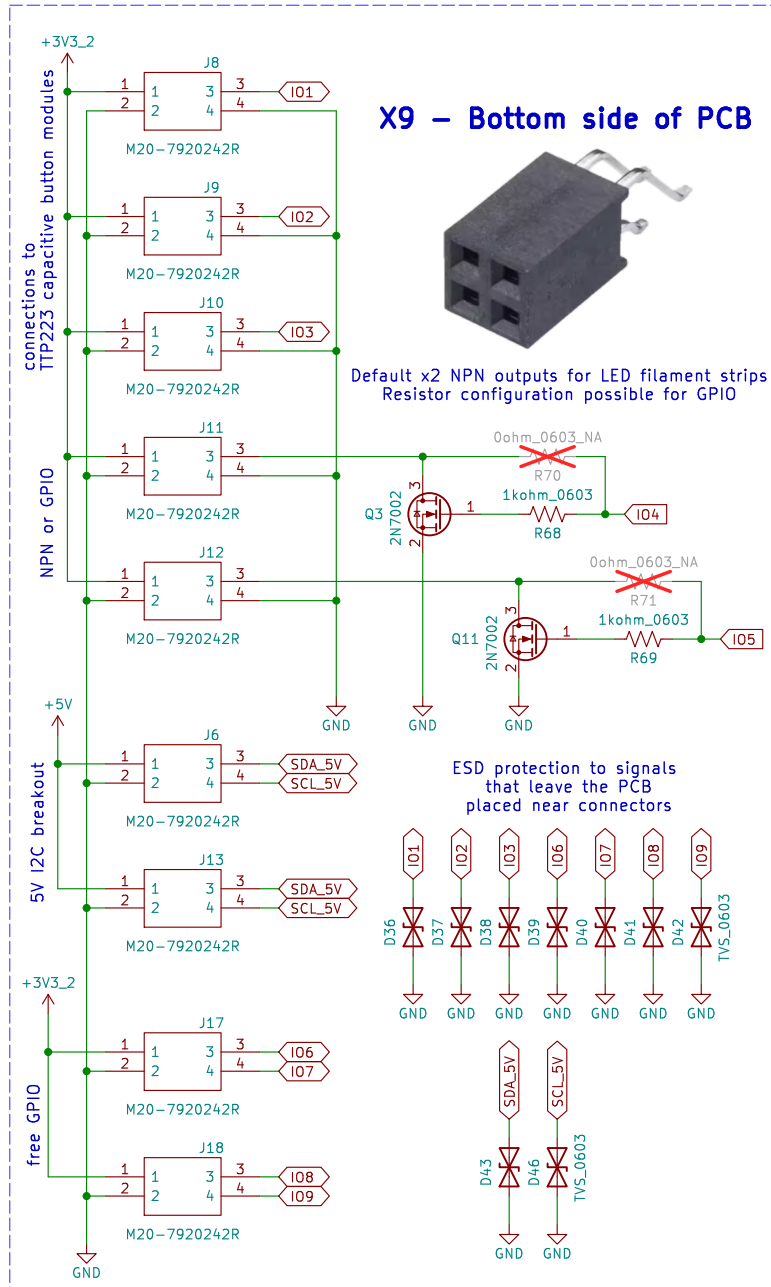
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Size: A4 Date: 2025-12-01

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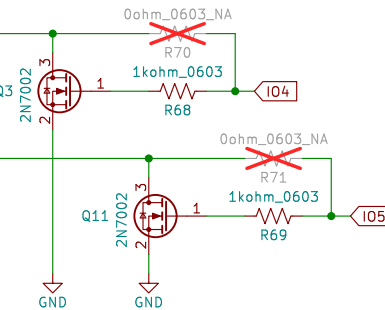
Rev: 1

Id: 8/9

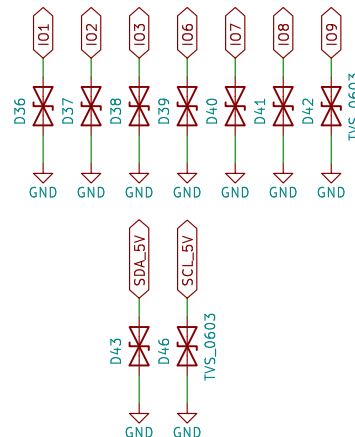


X9 – Bottom side of PCB

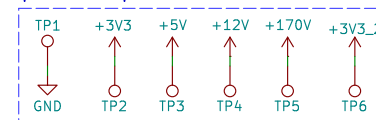
Default x2 NPN outputs for LED filament strips
Resistor configuration possible for GPIO



ESD protection to signals
that leave the PCB
placed near connectors



power test points



Sheet: /Connectors 2/
File: Connectors 2.kicad_sch

Size: A4	Date: 2025-12-01
KiCad E.D.A. 9.0.6	

Rev: 1
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